MOMENTS OF LEVEL WIDTHS IN CONDITIONED GALTON–WATSON TREES

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ABSTRACT. Let T_n be a simply generated random tree with n vertices, for instance a conditioned Galton–Watson tree, and let $l_k(T_n)$ be the number of vertices at distance k from the root. Janson conjectured that, when the generating function of the weights has a positive radius of convergence, $\mathbb{E}\, l_k(T_n)^r \to \mathbb{E}\, l_k(\widehat{T})^r \le \infty$ for all integers $r \ge 1$ and $k \ge 1$, where $\widehat{T}$ is the local limit of T_n, and that the same holds for non-integer $r > 0$. We show that if the offspring distribution is critical and has a finite moment of order $r+1$ for a real number $r \ge 1$, then the random variables $l_k(T_n)^r$, $n \ge 1$, are uniformly integrable for every k, and $\mathbb{E}\, l_k(T_n)^r \to \mathbb{E} Z_k^{r+1}$, where Z_k is the size of generation k of the unconditioned Galton–Watson process. Together with the cases that follow from Fatou's lemma, this proves the conjecture for every integer $r \ge 1$, and for every real $r > 0$ except in the case where $0 < r < 1$ and the offspring distribution is critical with infinite variance and a finite moment of order $r+1$. The proof bounds the density of the law of the first k generations of T_n with respect to that of the unconditioned tree by means of the local limit theorem.

1. INTRODUCTION

Let $w = (w_j)_{j \ge 0}$ be a sequence of nonnegative weights with $w_0 > 0$ and $w_j > 0$ for some $j \ge 2$. The simply generated random tree T_n is the random ordered rooted tree with n vertices whose probability of being a given tree t is proportional to $\prod_{v \in t} w_{d_t(v)}$, where $d_t(v)$ is the number of children of v in t; only those n are considered for which some tree with n vertices has positive weight. If w is a probability distribution, T_n is a Galton–Watson tree with offspring distribution w conditioned to have n vertices. We use the notation of Janson's survey [8], to which we refer for background.

Let $\Phi(z) = \sum_j w_j z^j$, let ρ be its radius of convergence, let $\Psi(t) = t\Phi'(t)/\Phi(t)$, and let $\nu = \lim_{t \uparrow \rho} \Psi(t) \le \infty$. We assume $\rho > 0$, which is equivalent to $\nu > 0$. If $\nu \ge 1$ let $\tau \in (0, \rho]$ be the solution of $\Psi(\tau) = 1$, and if $\nu < 1$ let $\tau = \rho$. Then $\pi_j = \tau^j w_j / \Phi(\tau)$ is a probability distribution with mean $\min(\nu, 1)$, and T_n has the law of a Galton–Watson tree with offspring distribution π conditioned to have n vertices [8, Section 4]. Let ξ be a random variable with distribution π and let $\sigma^2 = \operatorname{Var} \xi \le \infty$. If $\nu > 1$ then $\tau < \rho$ and ξ has finite exponential moments; in particular $\sigma^2 = \infty$ is possible only when $\nu \le 1$.

By [8, Theorem 7.1], T_n converges in distribution, in the topology of local convergence, to an infinite random tree $\widehat{T}$. When $\nu \ge 1$, the tree $\widehat{T}$ is the size-biased Galton–Watson tree of Kesten [9] (see also [10], [8, Section 5] and Abraham and Delmas [1]): it has an infinite path from the root, the spine, whose vertices have offspring distribution $\mathbb{P}(\hat{\xi} = j) = j\pi_j$, one child of each spine vertex, chosen uniformly, being the next spine vertex, while all other vertices have offspring distribution π, everything independent. When $0 < \nu < 1$, the tree $\widehat{T}$ has a vertex of infinite degree.

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For a rooted tree T let $l_k(T)$ be the number of vertices at distance k from the root. Janson [8, Section 21.1] showed that $l_k(T_n) \to l_k(\widehat{T})$ in distribution for every k; when $\nu \geq 1$ the limit is finite, and when $0 < \nu < 1$ the convergence holds in $[0, \infty]$ and $\mathbb{P}(l_k(\widehat{T}) < \infty) = \nu^k$. He also showed that $\mathbb{E}\, l_k(T_n) \to \mathbb{E}\, l_k(\widehat{T}) \leq \infty$ for every k; for $\nu \geq 1$ and $\sigma^2 < \infty$ this rests on the uniform integrability of $l_k(T_n)$, $n \geq 1$, see [7, Section 10], and for $\nu > 1$ the convergence was shown earlier by Meir and Moon [11]. For $r \geq 1$, Fatou's lemma gives $\mathbb{E}\, l_k(T_n)^r \to \mathbb{E}\, l_k(\widehat{T})^r = \infty$ when $0 < \nu < 1$ or $\mathbb{E}\xi^{r+1} = \infty$ [8, Section 21.1]. If r is an integer and $\mathbb{E}\xi^{r+1} < \infty$, the moments $\mathbb{E}\, l_k(T_n)^r$ are bounded uniformly in n by [7, Theorem 1.13], so that $\mathbb{E}\, l_k(T_n)^r \to \mathbb{E}\, l_k(\widehat{T})^r$ when $\mathbb{E}\xi^{r+2} < \infty$. Janson conjectured that the $(r+1)$-st moment suffices.[1]

Conjecture (Janson [8, Conjecture 21.1])**.** *For every integer $r \geq 1$ and every $k \geq 1$, if $\nu > 0$, then*

$$\mathbb{E}\, l_k(T_n)^r \to \mathbb{E}\, l_k(\widehat{T})^r \leq \infty. \tag{1.1}$$

This holds also for non-integer $r > 0$.

Janson conjectured that, when $\mathbb{E}\xi^{r+1} < \infty$, the variables $l_k(T_n)^r$, $n \geq 1$, are uniformly integrable, which together with the convergence in distribution of $l_k(T_n)$ yields (1.1). The following theorem proves this for every real $r \geq 1$.

Theorem 1.1. *Suppose that $\nu \geq 1$, let $r \geq 1$ be a real number, and suppose that $\mathbb{E}\xi^{r+1} < \infty$. Then for every $k \geq 1$ the random variables $l_k(T_n)^r$, $n \geq 1$, are uniformly integrable, and*

$$\mathbb{E}\, l_k(T_n)^r \to \mathbb{E}\, l_k(\widehat{T})^r = \mathbb{E} Z_k^{r+1} < \infty, \tag{1.2}$$

where Z_k is the number of vertices at distance k from the root in a Galton–Watson tree with offspring distribution π.

Corollary 1.2. *Let $\nu > 0$ and $k \geq 1$. Then (1.1) holds for every real $r \geq 1$. For $0 < r < 1$ it holds unless $\nu = 1$, $\sigma^2 = \infty$ and $\mathbb{E}\xi^{r+1} < \infty$. In particular the conjecture holds for every integer $r \geq 1$.*

In the remaining case $\nu = 1$, $\sigma^2 = \infty$, $0 < r < 1$, $\mathbb{E}\xi^{r+1} < \infty$, the limit $\mathbb{E}\, l_k(\widehat{T})^r = \mathbb{E} Z_k^{r+1}$ is finite, and the question whether $\mathbb{E}\, l_k(T_n)^r$ converges to it remains open; the proof below uses $\sigma^2 < \infty$ through the local limit theorem (Lemma 2.2).

The proof of Theorem 1.1 rests on an explicit formula for the law of the first k generations of T_n (Lemma 2.1), which is a special case of [8, Lemma 15.9], a lemma that Janson uses in the proof of [8, Theorem 7.1]. On trees with at least one vertex at distance k from the root, this law has a density W_n with respect to the law of the first k generations of the unconditioned Galton–Watson tree, and W_n converges to the number of vertices at distance k. The local limit theorem bounds W_n by a constant times this number as long as the first k generations contain at most half of the vertices, and the event that they contain more is controlled by the moment $\mathbb{E}\xi^{r+1}$.

Remark 1.3. For integer r the bound of [7, Theorem 1.13] is uniform in k: if $\mathbb{E}\xi^{r+1} < \infty$ then $\mathbb{E}\, l_k(T_n)^r = O(k^r)$ uniformly in $k \geq 1$ and $n \geq 1$. Addario-Berry, Devroye and Janson conjectured that this holds for all real $r \geq 1$ [2, Section 1]. At levels k of order $\sqrt{n}$ the widths $l_k(T_n)$ make up the profile of T_n, which after rescaling converges to the local time of a Brownian excursion, as conjectured by Aldous [3] and proved by Drmota and Gittenberger [4]. Under its hypotheses, Theorem 1.1 implies $\sup_n \mathbb{E}\, l_k(T_n)^r < \infty$ for each k and real $r \geq 1$, but the bound given by the proof grows faster than k^r, and that conjecture is not addressed here.

[1] In the arXiv version of [8] the conjecture is Conjecture 20.1, and Lemma 15.9, used below, is Lemma 14.9.

2. The first k generations of T_n

Throughout this section $\nu \geq 1$, so that $\mathbb{E}\xi = 1$. Since $\pi_0 > 0$, the variance σ^2 is positive. Let T be a Galton–Watson tree with offspring distribution π; then T_n is T conditioned on $|T| = n$, where $|T|$ is the number of vertices, and n ranges over the integers with $\mathbb{P}(|T| = n) > 0$. Let $h \geq 1$ be the span of ξ, the largest integer such that $\xi \in h\mathbb{Z}$ almost surely. Let $\xi_1, \xi_2, \dots$ be independent copies of ξ and $S_m = \xi_1 + \cdots + \xi_m$. For an ordered rooted tree t and $j \geq 0$, let $t^{[j]}$ be the subtree formed by the vertices at distance at most j from the root. Let $Z_j = l_j(T)$ be the generation sizes of T, and let $\mathcal{F}_j$ be the σ-field generated by $T^{[j]}$.

Fix $k \geq 1$. A *k-truncation* is a finite ordered rooted tree of height at most k. For a k-truncation t let $N(t)$ be the number of its vertices at distance less than k from the root, and let $l_k(t)$ be the number at distance k. Write

$$p(t) = \mathbb{P}(T^{[k]} = t) = \prod_v \pi_{d_t(v)},$$

the product over the vertices v of t at distance less than k from the root. Thus $\mathbb{E}f(T^{[k]}) = \sum_t f(t)p(t)$ for every $f \geq 0$, the sum over all k-truncations, and $l_k(T^{[k]}) = Z_k$, $N(T^{[k]}) = Z_0 + \cdots + Z_{k-1}$.

We use Dwass's formula [5]: for $l \geq 1$ and $m \geq 1$, the total number of vertices of a forest of l independent copies of T equals m with probability $\frac{l}{m}\mathbb{P}(S_m = m - l)$. For $l = 1$ it reads $\mathbb{P}(|T| = m) = \frac{1}{m}\mathbb{P}(S_m = m - 1)$; see also [8, Theorem 15.5].

Lemma 2.1. *Let t be a k-truncation with $N = N(t)$ and $l = l_k(t) \geq 1$, and let n satisfy $\mathbb{P}(|T| = n) > 0$. Then $\mathbb{P}(T_n^{[k]} = t) = p(t)W_n(t)$, where*

$$W_n(t) = \frac{n\, l\, \mathbb{P}(S_{n-N} = n - N - l)}{(n - N)\, \mathbb{P}(S_n = n - 1)} \quad \text{if } n > N, \qquad W_n(t) = 0 \quad \text{if } n \leq N. \tag{2.1}$$

Proof. Conditionally on $T^{[k]} = t$, the subtrees of T rooted at the l vertices at distance k from the root are independent copies of T, and $|T|$ is N plus their total number of vertices. By Dwass's formula, $\mathbb{P}(T^{[k]} = t,\ |T| = n) = p(t)\frac{l}{n-N}\mathbb{P}(S_{n-N} = n - N - l)$ for $n > N$, while for $n \leq N$ the left side is 0 because $l \geq 1$. Dividing by $\mathbb{P}(|T| = n) = \frac{1}{n}\mathbb{P}(S_n = n - 1)$ gives the claim. □

Lemma 2.1 is the case of [8, Lemma 15.9] in which the fixed subtree consists of the vertices of t at distance less than k from the root, with the numbers of children prescribed by t. Note that $W_n(t) = 0$ when $N < n < N + l$, since then $n - N - l < 0$.

Lemma 2.2. *Suppose that $\sigma^2 < \infty$.*

(i) *There is $C_1 < \infty$ such that $\mathbb{P}(S_m = j) \leq C_1 m^{-1/2}$ for all $m \geq 1$ and $j \in \mathbb{Z}$.*
(ii) *If $N \geq 1$ and $l \geq 0$ are integers with $N + l - 1 \in h\mathbb{Z}$, then $\mathbb{P}(S_{n-N} = n - N - l)/\mathbb{P}(S_n = n - 1) \to 1$ as $n \to \infty$ with $n \in 1 + h\mathbb{Z}$.*
(iii) *$c_2 = \inf\{n^{1/2}\mathbb{P}(S_n = n - 1) :\ n \geq 1,\ \mathbb{P}(S_n = n - 1) > 0\} > 0$.*

Proof. Since $\pi_0 > 0$, the variable ξ takes values in $h\mathbb{Z}$, and h is the maximal span. The local limit theorem for lattice distributions [6, Theorem 4.2.1] gives

$$\sup_{j \in h\mathbb{Z}} \Big| \frac{\sigma m^{1/2}}{h}\mathbb{P}(S_m = j) - \varphi\Big(\frac{j - m}{\sigma m^{1/2}}\Big)\Big| \to 0 \qquad (m \to \infty),$$

where φ is the standard normal density, and $S_m \in h\mathbb{Z}$. Since $\varphi \leq (2\pi)^{-1/2}$, (i) holds for $m \geq m_0$, and for $m < m_0$ we use $\mathbb{P}(S_m = j) \leq 1 \leq (m_0/m)^{1/2}$. In (ii) both $n - N - l$ and $n - 1$ lie in $h\mathbb{Z}$, and $l(n - N)^{-1/2} \to 0$, $n^{-1/2} \to 0$, so both probabilities equal $(1 + o(1))\, h\, (2\pi\sigma^2 n)^{-1/2}$. For

(iii), $\mathbb{P}(S_n = n-1) > 0$ forces $n \in 1 + h\mathbb{Z}$, and along $n \in 1 + h\mathbb{Z}$ we have $n^{1/2}\mathbb{P}(S_n = n-1) \to h(2\pi\sigma^2)^{-1/2} > 0$; the finitely many remaining values of n give positive terms. □

Lemma 2.3. *Let $0 \le j \le i$ and $s > 0$. Then $\mathbb{E}[Z_i^s \mid \mathcal{F}_j] \le C\, Z_j^s$ almost surely, where $C = \mathbb{E}Z_{i-j}^s \le \infty$ if $s \ge 1$ and $C = 1$ if $s \le 1$. Moreover $\mathbb{E}Z_g^s \le (\mathbb{E}\xi^s)^g$ for $s \ge 1$ and $g \ge 0$.*

Proof. Conditionally on $\mathcal{F}_j$, Z_i is the sum of Z_j independent copies of Z_{i-j}, namely the generation sizes at depth $i-j$ of the subtrees rooted at the vertices of generation j. For $s \ge 1$, Minkowski's inequality in L^s gives $\mathbb{E}[Z_i^s \mid \mathcal{F}_j]^{1/s} \le Z_j(\mathbb{E}Z_{i-j}^s)^{1/s}$. For $s \le 1$, Jensen's inequality for the concave function $x \mapsto x^s$ and $\mathbb{E}[Z_i \mid \mathcal{F}_j] = Z_j$ give $\mathbb{E}[Z_i^s \mid \mathcal{F}_j] \le Z_j^s$. Taking $i = g$ and $j = g-1$ in the first claim gives $\mathbb{E}Z_g^s \le \mathbb{E}\xi^s\, \mathbb{E}Z_{g-1}^s$, and the last claim follows by induction since $Z_0 = 1$. □

Lemma 2.4. *For every k-truncation t, $\mathbb{P}(\widehat{T}^{[k]} = t) = l_k(t)\, p(t)$. Consequently $\mathbb{E}f(l_k(\widehat{T})) = \mathbb{E}[Z_k f(Z_k)]$ for every measurable $f \ge 0$, and in particular $\mathbb{E}\, l_k(\widehat{T})^r = \mathbb{E}Z_k^{r+1}$ for $r > 0$.*

Proof. If $l_k(t) = 0$, both sides vanish, because the spine of $\widehat{T}$ has a vertex at distance k from the root. Otherwise let u be a vertex of t at distance k from the root, with ancestors $u_0, \dots, u_{k-1}$, where u_0 is the root; put $u_k = u$, and note that $d_t(u_i) \ge 1$ for $i < k$ since u_{i+1} is a child of u_i. The probability that $\widehat{T}^{[k]} = t$ and that the spine passes through u is

$$\prod_{i=0}^{k-1} \Big(d_t(u_i)\pi_{d_t(u_i)} \cdot \frac{1}{d_t(u_i)} \Big) \prod_v \pi_{d_t(v)} = p(t),$$

where the second product is over the vertices of t at distance less than k from the root other than $u_0, \dots, u_{k-1}$. Summing over the $l_k(t)$ choices of u gives the first claim. Then $\mathbb{E}f(l_k(\widehat{T})) = \sum_t f(l_k(t))l_k(t)p(t) = \mathbb{E}[Z_k f(Z_k)]$. □

Lemma 2.5. *Suppose that $\sigma^2 < \infty$. Then $\mathbb{P}(T_n^{[k]} = t) \to \mathbb{P}(\widehat{T}^{[k]} = t)$ as $n \to \infty$ for every k-truncation t. Consequently $l_k(T_n) \to l_k(\widehat{T})$ in distribution.*

Proof. If $p(t) = 0$, both sides are 0. If $l_k(t) = 0$, then $T_n^{[k]} = t$ forces $T_n = t$, so the left side is 0 for $n > N(t)$, and so is the right side. Suppose $p(t) > 0$ and $l = l_k(t) \ge 1$, and let $N = N(t)$. Every $d_t(v)$ with v at distance less than k from the root lies in $h\mathbb{Z}$, and the sum of these numbers is the number of vertices of t other than the root, which is $N + l - 1$. Hence $N + l - 1 \in h\mathbb{Z}$, and Lemma 2.2(ii) and $n/(n-N) \to 1$ give $W_n(t) \to l$. By Lemmas 2.1 and 2.4, $\mathbb{P}(T_n^{[k]} = t) = p(t)W_n(t) \to l\, p(t) = \mathbb{P}(\widehat{T}^{[k]} = t)$. The k-truncations form a countable set, and $\sum_t l_k(t)p(t) = \mathbb{E}Z_k = 1$, so by Scheffé's lemma the law of $T_n^{[k]}$ converges in total variation to that of $\widehat{T}^{[k]}$; in particular $l_k(T_n) \to l_k(\widehat{T})$ in distribution. □

Lemma 2.5 also follows from [8, Theorem 7.1], which does not assume $\sigma^2 < \infty$; the proof above uses only Lemmas 2.1, 2.2 and 2.4.

3. Proof of Theorem 1.1

Let $\nu \ge 1$, let $r \ge 1$ be real with $\mathbb{E}\xi^{r+1} < \infty$, and fix $k \ge 1$. Since $r + 1 \ge 2$, $\sigma^2 < \infty$, so Lemmas 2.2 and 2.5 apply, and $\mathbb{E}Z_j^{r+1} \le (\mathbb{E}\xi^{r+1})^j < \infty$ for every j by Lemma 2.3.

Bounds on the density. Let $C_3 = C_1/c_2$, with C_1 and c_2 from Lemma 2.2. Let t be a k-truncation with $N = N(t)$ and $l = l_k(t) \ge 1$, and let n satisfy $\mathbb{P}(|T| = n) > 0$. The two bounds below are trivial when $W_n(t) = 0$, so suppose $W_n(t) > 0$; then $n - N \ge l$ by (2.1). By (2.1) and Lemma 2.2(i),(iii),

$$W_n(t) \le \frac{n\, l\, C_1 (n-N)^{-1/2}}{(n-N)\, c_2 n^{-1/2}} = C_3\, l \Big(\frac{n}{n-N} \Big)^{3/2}, \tag{3.1}$$

and, since $n - N \geq l$,

$$W_n(t) \leq C_3\, n^{3/2}\, l^{-1/2}. \tag{3.2}$$

Uniform integrability. Let $K \geq 1$. Since $l_k(T_n) = l_k(T_n^{[k]})$, Lemma 2.1 gives

$$\mathbb{E}\big[l_k(T_n)^r;\ l_k(T_n) > K\big] = \sum_{t:\, l_k(t)>K} l_k(t)^r p(t) W_n(t) = \mathbb{E}\big[Z_k^r\, W_n(T^{[k]})\, \mathbf{1}\{Z_k > K\}\big]. \tag{3.3}$$

Let $N_k = N(T^{[k]}) = Z_0 + \cdots + Z_{k-1}$, and split the right side of (3.3) according to whether $N_k \leq n/2$.

On $\{N_k \leq n/2\}$, (3.1) gives $W_n(T^{[k]}) \leq 2^{3/2} C_3 Z_k$, so this part of (3.3) is at most

$$2^{3/2} C_3\, \mathbb{E}\big[Z_k^{r+1};\ Z_k > K\big]. \tag{3.4}$$

On $\{N_k > n/2\}$ some $j \in \{0, \ldots, k-1\}$ has $Z_j > n/(2k)$, and (3.2) gives $Z_k^r W_n(T^{[k]}) \leq C_3 n^{3/2} Z_k^s$ with $s = r - \frac{1}{2} \geq \frac{1}{2}$. Let $C_4 = \max\big(1, \max_{1\leq g\leq k} \mathbb{E} Z_g^s\big)$, which is finite because $s < r+1$. By Lemma 2.3 with $i = k$,

$$\begin{aligned}\mathbb{E}\big[Z_k^s\, \mathbf{1}\{Z_j > n/(2k)\}\big] &= \mathbb{E}\big[\mathbb{E}[Z_k^s \mid \mathcal{F}_j]\, \mathbf{1}\{Z_j > n/(2k)\}\big] \leq C_4\, \mathbb{E}\big[Z_j^s\, \mathbf{1}\{Z_j > n/(2k)\}\big] \\ &\leq C_4 \Big(\frac{2k}{n}\Big)^{3/2} \mathbb{E}\big[Z_j^{r+1};\ Z_j > n/(2k)\big],\end{aligned}$$

where the last step uses $Z_j^s = Z_j^{r+1} Z_j^{-3/2} \leq (2k/n)^{3/2} Z_j^{r+1}$ on $\{Z_j > n/(2k)\}$. Summing over j, this part of (3.3) is at most

$$\delta_n = C_3 C_4 (2k)^{3/2} \sum_{j=0}^{k-1} \mathbb{E}\big[Z_j^{r+1};\ Z_j > n/(2k)\big], \tag{3.5}$$

which does not depend on K and tends to 0 as $n \to \infty$ by dominated convergence, since $\mathbb{E} Z_j^{r+1} < \infty$.

Let $\varepsilon > 0$. Choose n_0 such that $\delta_n \leq \varepsilon$ for $n \geq n_0$, and $K_0 \geq n_0$ such that $2^{3/2} C_3 \mathbb{E}[Z_k^{r+1}; Z_k > K_0] \leq \varepsilon$. Let $K \geq K_0$. For $n < n_0$ we have $l_k(T_n) < n < K$, so the left side of (3.3) vanishes; for $n \geq n_0$ it is at most 2ε by (3.4) and (3.5). Since $\{l_k(T_n)^r > K^r\} = \{l_k(T_n) > K\}$, this gives $\sup_n \mathbb{E}[l_k(T_n)^r;\ l_k(T_n)^r > M] \leq 2\varepsilon$ for all $M \geq K_0^r$, so the variables $l_k(T_n)^r$, $n \geq 1$, are uniformly integrable.

The limit. By Lemma 2.5, $l_k(T_n)^r \to l_k(\widehat{T})^r$ in distribution. Uniform integrability and convergence in distribution imply convergence of the means, so $\mathbb{E}\, l_k(T_n)^r \to \mathbb{E}\, l_k(\widehat{T})^r$, which equals $\mathbb{E} Z_k^{r+1} \leq (\mathbb{E}\xi^{r+1})^k < \infty$ by Lemmas 2.4 and 2.3. This proves Theorem 1.1. □

4. Proof of Corollary 1.2

Let $\nu > 0$, $k \geq 1$ and $r > 0$. We use the convergence in distribution of $l_k(T_n)$ to $l_k(\widehat{T})$ from [8, Section 21.1], quoted in Section 1, and the following form of Fatou's lemma: if $X_n \to X$ in distribution in $[0, \infty]$, then $\liminf_n \mathbb{E} X_n^r \geq \mathbb{E} X^r$, because $x \mapsto \min(x, M)^r$ is bounded and continuous on $[0, \infty]$ for every $M > 0$.

Case $0 < \nu < 1$. Here $\mathbb{P}(l_k(\widehat{T}) = \infty) = 1 - \nu^k > 0$, so $\mathbb{E}\, l_k(\widehat{T})^r = \infty$, and Fatou's lemma gives $\mathbb{E}\, l_k(T_n)^r \to \infty$.

Case $\nu \geq 1$ *and* $\mathbb{E}\xi^{r+1} = \infty$. The event $\{Z_{k-1} \geq 1\}$ has positive probability, and on it Z_k is at least the number of children of the first vertex of generation $k-1$, which is independent of $\mathcal{F}_{k-1}$ and distributed as ξ. Hence $\mathbb{E} Z_k^{r+1} \geq \mathbb{P}(Z_{k-1} \geq 1) \mathbb{E}\xi^{r+1} = \infty$, so $\mathbb{E}\, l_k(\widehat{T})^r = \infty$ by Lemma 2.4, and Fatou's lemma gives $\mathbb{E}\, l_k(T_n)^r \to \infty$.

Case $\nu \geq 1$, $\mathbb{E}\xi^{r+1} < \infty$ *and* $r \geq 1$. This is Theorem 1.1.

Case $\nu \geq 1$, $0 < r < 1$ and $\sigma^2 < \infty$. Theorem 1.1 with exponent 1 shows that the variables $l_k(T_n)$, $n \geq 1$, are uniformly integrable, as was already shown in [7, Section 10]. Since $x^r \leq x$ for $x \geq 1$, for $K \geq 1$

$$\mathbb{E}\big[l_k(T_n)^r;\ l_k(T_n)^r > K\big] \leq \mathbb{E}\big[l_k(T_n);\ l_k(T_n) > K^{1/r}\big],$$

so the variables $l_k(T_n)^r$ are uniformly integrable as well, and (1.1) follows from the convergence in distribution.

These cases cover every $r \geq 1$. For $0 < r < 1$ the only case left out is $\nu \geq 1$, $\sigma^2 = \infty$, $\mathbb{E}\xi^{r+1} < \infty$, and $\sigma^2 = \infty$ forces $\nu = 1$ because ξ has finite exponential moments when $\nu > 1$. This proves Corollary 1.2. □

5. Computer checks

Lemma 2.1 and the inequalities of Lemma 2.3 were checked by computer. The script `code/checks.py` in the repository listed under Data availability uses only the Python standard library and runs in under a minute; its output is `code/checks.out`. It takes five critical offspring distributions: the geometric distribution $\pi_j = 2^{-j-1}$, the binomial distribution with parameters 2 and $\frac{1}{2}$, the uniform distribution on $\{0, 2\}$ (span 2), the distribution with $\pi_0 = \frac{1}{2}$, $\pi_1 = \pi_3 = \frac{1}{4}$, and the distribution with $\pi_0 = \frac{2}{3}$, $\pi_3 = \frac{1}{3}$ (span 3). For each of them and each n from 2 to 12 it enumerates all ordered trees with n vertices whose numbers of children lie in the support, computes the exact law of $T_n^{[k]}$ for $k = 1, 2, 3$, and compares it with $p(t)W_n(t)$ for every k-truncation t with $l_k(t) \geq 1$ and $N(t) + l_k(t) \leq n$, so that a truncation that does not occur must receive the value 0; a truncation with $l_k(t) = 0$, which is a whole tree of height less than k, is compared with $\mathbb{P}(T = t)/\mathbb{P}(|T| = n)$ if it has n vertices and with 0 otherwise; the probabilities $\mathbb{P}(S_m = j)$ are computed separately by convolution. All 35,302 comparisons agree, and Dwass's formula $\mathbb{P}(|T| = n) = \frac{1}{n}\mathbb{P}(S_n = n - 1)$ holds in every case. For the four distributions with finite support the script also checks $\mathbb{E}[Z_i \mid Z_j = m] = m$ and the inequalities of Lemma 2.3 for $m \in \{1, 2, 3, 4\}$ and $i - j \in \{1, 2, 3\}$, with $s \in \{1, 2, 3\}$ in exact arithmetic and $s \in \{\frac{1}{2}, \frac{3}{2}, \frac{5}{2}\}$ in floating point; all 336 checks pass. These computations test the identities and inequalities used in the proof; they are not part of it.

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This work was done by multiple AI agents, Claude (Anthropic) and Codex (OpenAI), connected to Math Brain [12] and working together inside the Brain framework [14], a shared, persistent memory in which the agents record claims, proofs, failed attempts and methods, retrieve one another's results, coordinate who works on what, and audit each other's work. For more on the details behind the Brain and Math Brain, see the Math Brain paper [13], the Math Brain repository [12] and the Brain framework repository [14].

Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics. Conjecture 21.1 of [8] was one of the open problems in probability on which the project's agents worked in September 2026. On 14 September 2026 Claude Fable 5.1 (Anthropic) found the proof of Theorem 1.1 and Corollary 1.2. On 12 October 2026 Claude Opus 5.5 (Anthropic), working in Claude Code, wrote the proofs in their present form, searched the literature, and wrote the computer checks and the manuscript. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Data availability

No data were used. The code is available at `https://github.com/yspennstate/level-width-moments`.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [12].

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